Trash Force Trio

Efficient Forest Robot System

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Introduction

The Trash Force Trio system addresses the challenges of forest maintenance by automating hazard detection, debris removal, and ecological monitoring. This three-robot solution is designed to enhance safety, reduce manual labor, and improve trail conditions over a 100 m × 100 m area, supporting efficient and continuous forest management operations.

01

Project Overview

Project Goals and Client Description

The project aims to develop an autonomous, coordinated robotic system to support forest management tasks. The key goals include improving trail safety, monitoring ecological data, and clearing obstacles. The client is focused on reducing maintenance delays, labor costs, and enhancing forest safety while providing an affordable and scalable solution.

Robot Models and Functions

The system consists of three robots: the Eco-Monitor collects soil moisture data; the Patrol Ranger detects trail hazards using sensors; and the Obstacle Guardian removes obstacles. Together, they operate autonomously, communicate within a 100 m radius, and work safely on uneven terrain, ensuring effective forest maintenance.

Expected Outcomes and Milestones

The project aims to deliver an autonomous robotic system that improves forest maintenance through soil monitoring, hazard detection, and obstacle removal. Milestones include prototype development, sensor integration, system testing, and final deployment. The expected outcome is a fully functional system that operates within a 100 m × 100 m area, communicates wirelessly, and enhances safety and efficiency in forest trail management.



02

Robot System Design and Operation

Sensors and Outputs of Each Robot

Each robot is equipped with specialized sensors tailored to its function. The Eco-Monitor uses soil moisture sensors and GPS for data collection. The Patrol Ranger employs ultrasonic and IR sensors to detect obstacles and uneven terrain. The Obstacle Guardian uses proximity sensors for obstacle detection and clearing mechanisms. Outputs include DC motors for movement and LEDs for hazard indication.

Autonomous Navigation and Obstacle Handling

The robots navigate autonomously using sensor data to avoid obstacles and manage difficult terrain. The Patrol Ranger reroutes upon detecting hazards under 20 cm and signals warnings with LEDs. The Eco-Monitor carefully reaches designated points for data gathering while avoiding obstacles. The system ensures continuous operation with minimal human intervention.

Robot Collaboration and System Integration

The three robots communicate continuously within a 100 m radius to coordinate tasks efficiently. This collaboration enables seamless monitoring, hazard detection, and debris clearing without overlap or interference. Integration supports system scalability and reliability, ensuring safe operation in uneven terrain and various light conditions.



Conclusions

The Trash Force Trio delivers a comprehensive, automated approach to forest maintenance. This coordinated robotic system meets all key operational requirements and offers significant advantages over manual methods, including reduced labor intensity and improved safety. Built from cost-effective components, it is scalable and practical, promising meaningful economic and ecological benefits for forest management agencies.

Thanks!

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